

Valuing Morbidity from Wildfire Smoke Exposure: A Comparison of Revealed and Stated Preference Techniques

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ABSTRACT. *Estimating the economic benefits of reduced health damages due to improvements in environmental quality continues to challenge economists. We review welfare measures associated with reduced wildfire smoke exposure, and a unique dataset from California's Station Fire of 2009 allows for a comparison of cost of illness (COI) estimates with willingness to pay (WTP) measures. The WTP for one less symptom day is estimated to be \$87 and \$95, using the defensive behavior and contingent valuation methods, respectively. These WTP estimates are not statistically different but do differ from a \$3 traditional daily COI estimate and \$17 comprehensive daily COI estimate. (JEL Q51, Q53)*

I. INTRODUCTION

A variety of environmental contaminants can negatively affect human health. Therefore government agencies, such as the U.S. Environmental Protection Agency (EPA), are tasked with protecting human health by reducing human exposure to contaminants in the air, water, and land. Branches of the EPA such as the National Center for Environmental Economics are responsible for analyzing the economic impacts, in other words, costs and benefits, of environmental regulations and policies. Quantifying these impacts can inform decision making by highlighting the trade-offs involved with policies aimed at improving environmental quality. However, the challenge of accurately monetizing the economic benefits of a reduction in human health damages associated with policies that effectively reduce pollution levels has remained pervasive in the economics literature, as well as the policy realm.

Freeman (2003) explains that improvements in environmental quality can benefit an affected individual by reducing some or all of the following adverse impacts: incurred medical expenses due to treating illness caused by pollution exposure; lost wages from the inability to work; expenditures and activities taken to defend oneself against the possible health effects associated with exposure; and the disutility associated with symptoms or lost leisure caused by illness. The lack of a monetary equivalent to some of these adverse effects complicates monetization of the full benefits of a reduction in pollution concentrations. In addition, an analyst cannot simply add together the monetary equivalent of changes in these components to arrive at the correct welfare measure of the benefits of reduced pollution concentrations or the health effects resulting from them. This is largely due to the fact that medical expenses are an exogenous outcome of illness, whereas defensive activities are chosen optimally by the individual (see Bockstael and McConnell 2007). Rather, there are measures based on these components that can be used to quantify, or approximate, these benefit estimates. Three approaches commonly used to monetize the benefits of improvements in human health associated with reductions in the concentrations of various pollutants are the defensive behavior method (DBM), the contingent valuation method (CVM), and the cost of illness (COI) approach.

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The contribution of this study is twofold. First, using unique primary data from the largest wildfire in Los Angeles County's modern history, we apply the DBM and CVM to estimate the willingness to pay (WTP) for a reduction in one wildfire smoke-induced symptom day. To our knowledge, these are the first WTP estimates for a reduction in symptom days associated with wildfire smoke exposure. Second, our data allow for a comparison of estimates across all three common approaches used to monetize the benefits of reduced morbidity. To statistically test for a significant difference in the three estimates, we test for overlapping confidence intervals and, where appropriate, carry out a complete combinatorial test. In addition, the study results shed light on WTP:COI calibration factors for the health damages associated with wildfire smoke specifically.

II. THEORETICAL FRAMEWORK

Defensive Behavior Method

The DBM is a revealed preference approach used in the field of health and environmental economics. The method originates from a health production function first outlined by Grossman (1972), with extensions to the model undertaken by Cropper (1981) and Harrington and Portney (1987). The general idea of the DBM in the health production function framework is that if an individual experiences some negative health output, such as a number of days with symptoms, it enters into his utility function as follows:

$$U = U(X, L, S), \quad [1]$$

where X represents consumption of a composite market good, L represents leisure time, and S represents symptom days. It can be assumed that utility is increasing in consumption of X and L and decreasing in S . This health output S is not exogenous but rather "produced" by the individual according to a health production function:

$$S = S(P, D, Z), \quad [2]$$

where P represents exposure to a pollutant and Z represents a set of exogenous factors that

can affect symptom days, such as lifestyle and demographic factors, as well as health status prior to exposure. Further, the individual can choose to invest in the production of his health by taking defensive actions, labeled D . These include actions that serve to decrease symptom days by decreasing exposure to a given pollutant. Examples include evacuating a polluted area, staying indoors, using a home air cleaner to reduce indoor pollution levels, or using preventative medications such as a prescribed inhaler. Symptom days are increasing in P and decreasing in D . Following closely the framework in Dickie (2003), the individual is also subject to a full income budget constraint as follows:

$$\bar{I} + w[T - L - S] = X + p_d D + M(S), \quad [3]$$

where $\bar{I}$ represents nonlabor exogenous income, w represents labor income, and the individual is assumed to allocate his total time available T between work, leisure, and time spent sick. The price of X is normalized to 1, defensive actions have an associated cost of p_d , and the individual may also incur medical costs as a function of the possible illness experienced, represented by $M(S)$. This constraint can be rewritten as

$$\bar{I} + wT = X + wL + p_d D + M(S) + wS. \quad [4]$$

Therefore, the individual's total budget is spent on the consumption of goods, leisure, defensive actions, medical expenditures, and indirect costs of time lost due to illness. The individual's costs of illness are represented by the last two components of [4]:

$$COI(S) = M(S) + wS. \quad [5]$$

These costs represent an exogenous outcome of illness. Following Dickie (2003), this utility maximization problem can be solved in stages, where the individual first chooses defensive actions to minimize defensive expenditures ($p_d D$), subject to producing a given amount of symptom days, resulting in the following first-order conditions:

$$\begin{aligned} p_d - \pi(\partial S(\cdot)/\partial D) &= 0, \\ S^0 - S(\cdot) &= 0, \end{aligned} \quad [6]$$

where S^0 is the given amount of symptom days and π is the Lagrangean multiplier. These first-order conditions can be solved to obtain values of D and π as a function of prices, symptom days, pollution levels, and any other exogenous factors that minimize defensive expenditures while achieving symptom days S^0 . These can be used to define the defensive expenditure function (Bartik 1988) as follows:

$$DE(p_d, S^0, P, Z) = p_d D^0. \quad [7]$$

Utilizing first-order conditions and the envelope theorem, which tells us that the marginal cost of symptom days equals the Lagrangian multiplier π , we have

$$(\partial DE / \partial S) = \pi = p_d / (\partial S / \partial D) < 0. \quad [8]$$

In the second stage, the individual chooses the amount of symptom days and consumption of goods and leisure to maximize utility. This can be solved by plugging the defensive expenditure function into the full income budget constraint as follows:

$$\bar{I} + wT = X + wL + DE(p_d, S^0, P, Z) + M(S) + wS. \quad [9]$$

The utility maximization problem becomes

$$\begin{aligned} \text{Maximize } U &= U(X, L, S, P, D, Z) \\ \text{s.t. } \bar{I} + wT &= X + wL + DE(p_d, S^0, P, Z) + M(S) + wS \end{aligned} \quad [10]$$

The first-order conditions can then be obtained, with the one related to the individual's choice of symptom days being

$$(\partial U / \partial S) - \lambda(\partial DE / \partial S) - \lambda(dM/dS) - \lambda w = 0, \quad [11]$$

which can be rearranged as

$$-(\partial DE / \partial S) = -(\partial U / \partial S) / \lambda + (dM/dS) + w. \quad [12]$$

Finally, to estimate the marginal value of a change in illness dS , which is the focus of this paper, equations [8] and [12] can be used to show that

$$\begin{aligned} -(\partial DE / \partial S)dS &= [-(\partial U / \partial S) / \lambda + (dM/dS) + w]dS \\ &= -[p_d / (\partial S / \partial D)]dS. \end{aligned} \quad [13]$$

By utility maximization, the marginal benefit of reduced time spent sick, that is, the WTP for one less symptom day, must equal the marginal cost of reduced sick time. Since defensive actions are an input to the production of the negative health output, it follows from production theory that the marginal cost of reduced symptom days will equal the ratio of the full price of any of these defensive actions to its marginal product.

Contingent Valuation Method

Unlike the DBM, which looks at the defensive actions individuals take to arrive at a measure of the economic value of a decrease in symptom days, the CVM is a stated preference approach used to estimate this value. In a CVM framework, individuals are asked directly about the value they place on a specific change in a nonmarket good, which in this case would be a decrease in the number of symptom days experienced as a result of exposure to some pollutant. Following the DBM framework, the individual can solve his dual problem of minimizing expenditures subject to a given level of utility, say, u^* . This expenditure minimization problem can be solved to obtain the minimum expenditure function as follows:

$$e = e(p_d, P, Z, S^0, u^*). \quad [14]$$

This is the minimum expenditure required to remain at utility level u^* given symptom days S^0 , prices p_d , a particular pollution level P , and exogenous characteristics of the individual Z . The WTP for a reduction in symptom days from S^0 to S^1 can be expressed as

$$WTP = e(p_d, P, Z, S^0, u^*) - e(p_d, P, Z, S^1, u^*). \quad [15]$$

This shows the maximum amount of money the individual would pay to enjoy fewer symptom days while maintaining the same level of utility.

COI Approach

Given the complexities involved with empirically estimating WTP, agency economists and health professionals frequently rely on a

simple COI approach to capture the benefits of changes in morbidity resulting from reduced pollution concentrations. This approach focuses on capturing the direct and indirect resource costs associated with a pollution-induced illness. These costs typically include expenditures on medications and medical care used to treat the illness, the opportunity cost of time spent obtaining medical care, as well as the economic cost of lost wages resulting from time spent sick. The benefits of reduced pollution are measured as the associated reduction in these resource costs (Dickie 2003).

While a COI estimate captures resource costs associated with a pollution-related illness, equation [12] demonstrates theoretically why the COI approach often underestimates the true individual value of a reduction in illness caused by pollution. The marginal benefit the individual receives from reduced time spent sick is the sum of the monetary equivalent of disutility avoided ($-(\partial U/\partial S)/\lambda$), the individual's savings on medical expenditures (dM/dS), and the marginal value of additional time available for work and leisure (w). Monetization of the savings in costs of illness will fail to capture the disutility avoided as a result of improved health. A similar theoretical relationship is derived by Freeman (2003), who explains that for a reduction in illness, marginal WTP is the sum of the COI and the monetary equivalent of the lost utility. However, under very particular circumstances in which exposure to the pollutant does not cause disutility, and the individual cannot take defensive actions to mitigate the negative effects of changes in pollution, the COI will equal the correct welfare measure for a change in pollution concentrations. See Bockstael and McConnell (2007) for the derivation of this case.

The shortcomings of the COI approach in the more general setting where there is disutility of illness are highlighted by the EPA's National Center for Environmental Economics statement that "the practical problem [with this approach] is that unit costs for morbidity effects usually are measured in terms of avoided medical outlays and wages, which likely underestimate what people would be willing to pay to avoid the adverse health effects in question" (USEPA 2010b). However,

there are various pollutants for which monetized COI estimates remain the only available source of information on the potential benefits of reduced pollution. In these circumstances, policy makers have little choice but to utilize them in damage assessments, whether or not the assumptions necessary for them to represent a correct welfare measure hold. The EPA reports that "many important morbidity effects are poorly studied from the WTP perspective. The COI approach is much more common in valuing chronic illness" (USEPA 2010b).

III. A REVIEW OF THE LITERATURE

The DBM has been used to calculate the value of a reduction in a number of air and water pollutants and the health damages associated with exposure to them, including but not limited to sulfur dioxide (Cropper 1981; Joyce, Grossman, and Goldman 1989), nitrogen dioxide (Dickie and Gerking 1991a), carbon monoxide (Dickie and Gerking 1991a), ozone (Dickie et al. 1986; Gerking and Stanley 1986; Dickie et al. 1987; Dickie and Gerking 1991a, 1991b), and contaminated water supplies (Harrington, Krupnick, and Spofford 1989; Um, Kwak, and Kim 2002; Dasgupta 2004). In addition, numerous studies have applied the CVM to estimate the WTP to avoid the health damages associated with various pollutants (Rowe and Chestnut 1985; Dickie et al. 1986; Tolley et al. 1986; Dickie et al. 1987; Chestnut et al. 1996; Alberini and Krupnick 2000).

A number of the above studies have looked at the relationship between COI estimates and WTP values for a reduction in health effects from exposure to a pollutant, and the majority of empirical findings support theoretical predictions that costs of illness underestimate WTP. For instance, Rowe and Chestnut (1985) interviewed a panel of asthmatics in Glendora, California, and found that CVM WTP estimates for reductions in the severity of asthma symptoms were 1.6 to 3.7 times the comparable COI estimates. Dickie and Gerking (1991b) interviewed residents in Glendora and Burbank, California, and found that WTP for decreased ozone levels exceeded medical expenses by a factor of 2 to 4. Chestnut et al.